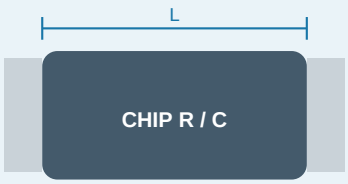
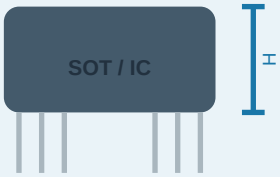
SMD PACKAGE SIZE REFERENCE GUIDE

A visual field guide to common surface-mount package families for ICs, diodes, resistors, capacitors, transistors, MOSFETs, LEDs, and related parts.

SAMPLE PACKAGE PICTURE - simplified views, not to scale



TOP VIEW: length L x width W



SIDE VIEW: height H

Pitch = center-to-center lead spacing (ICs)

Dimensions in this guide: package body L x W x H; pin pitch shown separately. PCB pads are not included.

How to read a package size: L x W x H is the component body length, width, and maximum height in millimeters. For gull-wing and leaded parts, lead span can be wider than the body. For leadless parts, terminal dimensions are separate. Pin pitch is center-to-center spacing.

Important: Package names are not perfectly uniform across brands. “Typical” dimensions below are nominal identification values, not production limits. Use the exact part datasheet for tolerances, pinout, orientation, thermal pad, and recommended PCB land pattern. IPC-7351 addresses land patterns, not a universal component-body catalog. [1]

Quick package family map

Component group	Common package families
Resistors / capacitors / small LEDs	Chip sizes: 01005 to 2512; MELF and arrays also common
Diodes / rectifiers / TVS	SOD-323/123/523; SMA/SMB/SMC; SOT-23; DFN
Small transistors / MOSFETs	SOT-23, SOT-323, SOT-89, SOT-223, DFN; power DPAK
ICs / logic / analog / memory	SOIC, TSSOP, SSOP, QFN/DFN, LQFP, BGA, WLCSP
Power / protection / RF / opto	DPAK/D2PAK, PowerSO, SMA/SMB/SMC, SOD, SOP-4, RF leadless

1. Chip passives and small indicators

Resistors, MLCC capacitors, ferrite beads, and many LEDs share the familiar two-terminal chip-size codes. Imperial and metric codes can collide: EIA 0402 is 1.0 x 0.5 mm, while metric 0402 is 0.4 x 0.2 mm. Always confirm which numbering system is used. Murata lists the EIA/JIS mappings and representative MLCC dimensions. [2]

EIA / inch code	JIS / metric code	Typical body L x W (mm)	Typical H (mm)	Notes
01005	0402 metric	0.40 x 0.20	0.13-0.23	Very small; check assembly capability
0201	0603 metric	0.60 x 0.30	0.20-0.40	Common miniature passive
0402	1005 metric	1.00 x 0.50	0.30-0.60	Very common; metric/EIA name confusion
0603	1608 metric	1.60 x 0.80	0.35-0.80	General purpose
0805	2012 metric	2.00 x 1.25	0.50-1.10	General purpose, larger ratings
1206	3216 metric	3.20 x 1.60	0.55-1.40	Power and voltage vary by series
1210	3225 metric	3.20 x 2.50	0.60-2.30	Wider body, common capacitor size
1812	4532 metric	4.50 x 3.20	0.80-3.00	Bulkier passive / protection
2010	5025 metric	5.00 x 2.50	0.60-1.20	Often resistor size code
2512	6432 metric	6.30 x 3.20	0.60-1.20	Power resistor and shunt use

Other passive package shapes

Family	Representative dimensions	Where used / caution
MELF / MiniMELF	Cylindrical 1.4-5.8 mm long; 1.0-2.2 mm diameter	Resistors and diodes; orientation can matter
Resistor networks / arrays	Common bodies about 2.0-5.1 mm long; 2-8+ terminals	Check pin count, internal circuit, and terminal pitch
Tantalum / polymer capacitor	A / B / C / D case families; roughly 3.2-7.3 mm long	Case code is not the same as chip-resistor code
Power inductor	Broad range, often 2-15+ mm footprint	Size, height, saturation current, and shielding vary widely
Crystal / oscillator	Common 2.0 x 1.6, 2.5 x 2.0, 3.2 x 2.5 mm bodies	Package body only; pad and keepout are device-specific

Passive-size decoding: The first two digits usually encode length and last two encode width in hundredths of an inch for EIA chip codes (e.g., 0603 ≈ 0.06 x 0.03 in). This is approximate; use the metric body dimensions for design. Capacitor thickness and resistor power rating vary by manufacturer and series. [2]

2. Diodes, LEDs, and discrete semiconductors

The same outline can contain different electrical functions. SOD/SMA names describe package families, not whether a part is a signal diode, Schottky, Zener, TVS, or rectifier. Check polarity markings and pin configuration in the part datasheet. Diodes Incorporated maintains outline drawings for many SOD, SOT, and DFN variants; Vishay lists typical rectifier package dimensions. [3][4]

Package	Typical role	Typical body L x W x H (mm)	Pitch (mm)	Common use
SOD-523	Small diode	1.6 x 0.8 x 0.6	-	Tiny signal / Zener
SOD-323	Small diode	1.7 x 1.25 x 0.95	-	Signal diode / low-power Zener
SOD-123	Small diode	2.65 x 1.6 x 1.1	-	Common signal, Schottky, Zener
SOD-123F	Flat-lead diode	2.6 x 1.6 x 0.7	-	Low profile; variants differ
SOT-23 / SOT-23-3	3-lead transistor	2.9 x 1.3 x 1.0	0.95	BJT, MOSFET, dual diode/regulator
SOT-23-5 / -6	5/6-lead device	2.9 x 1.6 x 1.1	0.95	ICs, dual FETs, transistor arrays
SOT-323 / SC-70	3-lead small outline	2.0 x 1.25 x 0.9	0.65	Small signal transistor/diode
SOT-363 / SC-70-6	6-lead small outline	2.0 x 1.25 x 0.9	0.65	Dual transistor / array
SOT-89	Power transistor	4.5 x 2.5 x 1.5	1.5	BJT / regulator / small power FET
SOT-223	Power outline	6.5 x 3.5 x 1.6	2.3 typical	Regulator / transistor / MOSFET; large tab
SMA / DO-214AC	Rectifier / TVS	5.1 x 2.7 x 2.1	—	Common 1 A-class rectifier outline
SMB / DO-214AA	Rectifier / TVS	5.4 x 3.6 x 2.3	—	Larger diode / protection device
SMC / DO-214AB	Rectifier / TVS	7.9 x 5.9 x 2.3	—	Higher power protection / rectifier
SMF / DO-219AB	Slim diode	3.7 x 1.8 x 1.0	—	Compact TVS / rectifier family
MicroSMP / DO-219AD	Tiny diode	2.5 x 1.3 x 0.65	—	Low profile / TVS families

LEDs and photoelectric parts

Family	Common size / outline	Representative body size	Check
Chip LED	0201 / 0402 / 0603 / 0805	0.6x0.3 through 2.0x1.25 mm	Check anode/cathode and lens height
PLCC-2 / PLCC-4	2 or 4 pads	~3.2x2.8 to 5.0x5.0 mm	Common indicator and RGB LED
SMD IR / photodiode	SOD, side-view, leadless	Varies by optics and window	Optical face and pinout matter

Dimension note: SOD-123 is a representative outline. For example, Diodes Inc. specifies a typical 2.65 mm body length, 1.55 mm body width, and 1.05 mm height (with min/max limits); another maker's drawing may differ slightly. SMA/SMB/SMC values above are rounded representative outlines. [3][4]

3. IC packages with leads

A package may be sold in multiple pin counts and body widths. The dimensions below are representative body dimensions; “span” means the overall outside-to-outside lead dimension when useful. Pitch is the center spacing of adjacent pins along a row. TI’s package finder provides exact body, pitch, height, and drawings for its package variants. [5]

Package family	Typical pins	Representative body L x W x H (mm)	Pitch (mm)	Typical applications
SOT-23-5 / 6	5-6	2.9 x 1.6 x 1.1	0.95	Small analog IC, logic, regulator
SOIC narrow	8-16	~4.9 x 3.9 x 1.75	1.27	Op amp, memory, controller
SOIC wide	8-16	~10.3 x 7.5 x 1.75	1.27	Wide-body isolation / power
SOP / SSOP	8-28+	~3-10 x 3-8 x 1.5-2.0	1.0 / 0.65	Compact logic and analog
TSSOP	8-56	~3-12 x 3-8 x 1.0-1.2	0.65 / 0.5	Thin, higher pin density
MSOP / VSSOP	8-16	~3.0 x 3.0 x 1.1	0.65 / 0.5	Small analog and power control
QSOP	16-28	~6-10 x 3.8 x 1.5	0.635	Shrink small outline
LQFP / TQFP	32-256	~7-28 square x 1.0-1.6	0.8 / 0.5 / 0.4	MCUs, DSPs, ASICs
PLCC	20-84	~9-30 square x 3.0-4.6	1.27	J-lead, socketable variants
PowerSO / HSOP	8-28	~5-16 x 5-11 x 1.7-3.0	0.65-1.27	Power IC, driver, heat spreading

Common shorthand and aliases

Short name	Meaning / note
SOIC	Small outline IC; also SO / SOP depending on vendor
TSSOP	Thin shrink small-outline; pin count, body width, and pitch define variant
MSOP / VSSOP	Miniature / very thin small-outline; names overlap across vendors
LQFP / TQFP	Low-profile / thin quad flat package; lead count and pitch vary widely
PLCC	Plastic leaded chip carrier; J-shaped leads

Read the part number suffix: manufacturers often encode package and tape/reel options at the end of an IC order code. Similar device names can be offered in several outlines. Match the full ordering code to the outline drawing before making a PCB footprint.

4. Leadless, array, power, and special packages

Leadless packages use pads under or around the body rather than visible gull-wing leads. The exposed thermal pad is an electrical and thermal feature; its dimensions and via guidance must come from the individual datasheet. BGA and wafer-level packages are specified by body size, ball/pad count, and pitch rather than gull-wing lead pitch. [5][6]

Package family	Typical terminals	Representative body L x W x H (mm)	Pitch / spacing	Typical use
DFN / SON	6-16	~2-6 x 2-6 x 0.5-1.0	0.4-0.8	Small leadless; optional exposed pad
QFN / VQFN	16-100+	~3-12 square x 0.5-1.0	0.4-0.8	Logic, MCU, RF, power; exposed pad common
WSON	6-16	~2-5 x 2-5 x 0.5-0.8	0.4-0.65	Very thin small-outline no-lead
LGA	varies	~2-15+ x 2-15+ x 0.5-2	0.4-1.0	Sensors, RF, modules; land grid
BGA / FBGA	25-1000+	~3-40 square x 0.5-3	0.5-1.27 ball pitch	Processors, memory, FPGA
WLCSP / CSP	4-200+	die-sized, often 0.5-5 square	0.3-0.8 ball pitch	Tiny mobile / power / sensor IC
DPAK / TO-252	3-7	~6.5 x 6.1 x 2.3	2.28-2.54	Power MOSFET, regulator, rectifier
D2PAK / TO-263	3-7	~10 x 15.4 x 4.4	2.54-5.08	Higher power MOSFET / rectifier
PowerPAK / LFPAK	varies	~3-10 x 3-11 x 0.8-2.5	vendor-specific	Low-resistance FET; footprint varies
SMA/SMB/SMC	2	5.1x2.7x2.1 / 5.4x3.6x2.3 / 7.9x5.9x2.3	—	Rectifiers, TVS and surge protection

Small cross-reference: package family by component

Part type	Common SMD families (examples, not a complete catalog)
Resistor	Chip (01005-2512), MELF, array, power resistor
Capacitor	Chip MLCC, tantalum/polymer case, aluminum can, feedthrough
Inductor / ferrite	Chip, molded/shielded power inductor, bead, common-mode choke
Diode / TVS / rectifier	SOD, SMA/SMB/SMC, DFN, bridge and power leadless packages
BJT / small MOSFET	SOT-23, SOT-323, SOT-363, SOT-89, SOT-223, DFN
Power MOSFET / regulator	DPAK, D2PAK, PowerSO, PowerPAK, LFPAK, QFN
IC / logic / analog / MCU	SOIC, TSSOP, QFN/DFN, LQFP, BGA, WLCSP
LED / optocoupler / sensor	Chip LED, PLCC, SOD/SOT, SOP-4, LGA, QFN

Other SMD parts found on boards

Group	Examples	Key check
Board connectors	FPC/FFC, board-to-board, USB, card, RF coax	Body sizes and contact pitch are connector-specific; inspect mating side and retention features.
Switches / buttons	Tact switch, slide switch, encoder	Body and actuator dimensions matter as much as solder terminals.
Fuse / protection	Chip fuse, resettable PTC, ESD array	Check current rating, voltage rating, polarity, and thermal limits.
Relays / optocouplers	SMD relay, SOP-4/6/8 optocoupler	Often taller and larger; observe isolation and keepout requirements.

5. Use this guide safely

A package label narrows the search; it does not uniquely identify the part. Many outlines have competing names, variants, or manufacturer-specific body dimensions. Package dimensions below are rounded common examples for recognition and rough planning. Use the manufacturer mechanical drawing before ordering boards or footprints.

PCB footprint checklist

Step	Check
1	Confirm the full orderable part number and package code.
2	Open its package drawing: body size, lead/pad size, tolerances, height, pin-1 mark, polarity.
3	Read the pin map and numbering orientation from the top-view drawing.
4	Copy or derive PCB land pattern from the datasheet or an applicable IPC-7351 land-pattern recommendation.
5	For thermal/power parts, follow exposed-pad, copper, via, and heat-dissipation guidance.
6	Check stencil, solder mask, courtyard, and assembly capabilities for very small parts.

Units and abbreviations

Term	Meaning
L x W x H	Body length x body width x maximum height, millimeters unless stated
Pitch	Center-to-center spacing between adjacent terminals
EIA / imperial chip code	Common resistor/capacitor size code such as 0603
JIS / metric code	Metric outline code such as 1608 for 1.6 x 0.8 mm
DFN / QFN / SON / WSON	Leadless package families; terminal geometry is underside/edge-specific
BGA / WLCSP	Ball array / wafer-level chip-scale package

References and package outline sources

[1] IPC-7351 - Generic requirements and guidance for PCB surface-mount land patterns. Open source

[2] Murata MLCC size table - EIA/JIS size mapping and representative body dimensions for chip capacitors. Open source

[3] Diodes Incorporated package outlines - Package drawings and pad layouts for SOD, SOT, DFN and related packages. Open source

[4] Vishay eSMP dimensions - Representative SMA, SMB, SMC, SMF, MicroSMP, DPAK and power package sizes. Open source

[5] Texas Instruments package finder - Manufacturer package dimensions, pitches, heights, and drawings for IC packages. Open source

[6] NXP package search - Package family descriptions, JEDEC outline references, dimensions, and drawings. Open source

Prepared as a practical overview of widely used SMD families. Dimensions are rounded typical identification values and may differ by vendor, tolerance, lead style, and subfamily. Source pages accessed 28 October 2026.